

CRF: K25 and the Two-Layer Pinning

K21 pins $n = 8$ from geometry alone. K20 then derives ε_0 from e . ε_0 cancels in K21. The golden ratio selects $n = 8$. Euler's e fixes the noise floor.

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Abstract

Two structural findings emerge from analysis of K20 and K21 (CRF K20–K24).

K21 is ε_0 -independent. The identity $T_{\text{avg}} \cdot \sigma_{\Psi} / \varphi_{\text{var}} = \sqrt{\varphi}$ expands to:

$$\frac{2D_0(n+2)}{(n-1)\sqrt{2n+3}\varphi_{\text{var}}} = \sqrt{\varphi}$$

The ε_0 terms cancel exactly. K21 is a *pure function of n* that equals $\sqrt{\varphi}$ at $n = 8$. The equation $f(n) = \sqrt{\varphi}$ has its unique positive solution at $n = 8.00045$ — selecting the integer $n = 8$ by proximity.

K20 derives ε_0 . Given $n = 8$ and Euler’s e , the identity $K^*T_{\text{avg}}/F = e$ reduces to a quadratic equation in ε_0 whose positive root is $\varepsilon_0 = 0.007550$ — matching the EIC/SINDy value to 0.005%.

Two-layer pinning:

1. K21 selects $n = 8$ from pure geometry ($\sqrt{\varphi}$).
2. K20 derives ε_0 from $n = 8$ and Euler’s e .

Together, K20 and K21 constitute an over-determination of the two CRF primitives (n, ε_0) from mathematical constants (φ, e) alone.

K25 (0.004%): $K^*\varphi_{\text{var}}/(F\sigma_{\Psi}) = e/\sqrt{\varphi}$ — the ratio of K20 to K21, an algebraic corollary.

1 K21 is ε_0 -Independent

1.1 The cancellation

K21 states $T_{\text{avg}} \cdot \sigma_{\Psi} / \varphi_{\text{var}} = \sqrt{\varphi}$. Substituting the derived formulas:

- $T_{\text{avg}} = 2nK^*/[(n-1)\varepsilon_0] = 2D_0/[(n-1)\varepsilon_0]$
- $\sigma_{\Psi} = \varepsilon_0(n+2)/\sqrt{2n+3}$
- $\varphi_{\text{var}} = 1/2 - 1/n^2 = (n^2 - 2)/(2n^2)$

$$T_{\text{avg}} \cdot \sigma_{\Psi} = \frac{2D_0}{(n-1)\varepsilon_0} \cdot \frac{\varepsilon_0(n+2)}{\sqrt{2n+3}} = \frac{2D_0(n+2)}{(n-1)\sqrt{2n+3}}$$

K21 is ε_0 -free

$$\frac{T_{\text{avg}} \cdot \sigma_{\Psi}}{\varphi_{\text{var}}} = \frac{2D_0(n+2)}{(n-1)\sqrt{2n+3}\varphi_{\text{var}}} = \frac{4n^3(1 - e^{-1/n})(n+2)}{(n-1)\sqrt{2n+3}(n^2 - 2)} \equiv f(n)$$

The ε_0 factors cancel exactly. K21 is a **pure function of n** depending only on the SO(3)-derived mode count.

At $n = 8$: $f(8) = 1.27208 \approx \sqrt{\varphi} = 1.27202$ (gap 0.004%).

1.2 $\sqrt{\varphi}$ selects $n = 8$

The equation $f(n) = \sqrt{\varphi}$ has a unique positive solution $n^* = 8.00045$, whose nearest integer is $n = 8$. No other integer $n \in \{3, 4, \dots, 12\}$ satisfies K21 within 10%:

n	$f(n)$	$\sqrt{\varphi}$	Gap
6	1.612	1.272	26.7%
7	1.414	1.272	11.1%
8	1.272	1.272	0.004%
9	1.165	1.272	8.4%
10	1.080	1.272	15.1%

The golden ratio $\sqrt{\varphi}$ is the $n = 8$ eigenvalue of the CRF wave dynamics.

2 K20 Derives ε_0 from $n = 8$ and e

2.1 The derivation

K20: $K^*T_{\text{avg}}/F = e$. Substituting $K^*T_{\text{avg}} = 2D_0^2/[n(n-1)\varepsilon_0]$ and $F = (1 + \varphi_{\text{var}})(1 + \psi_{\text{grad}})$ where $\psi_{\text{grad}} = C\varepsilon_0$ with $C = (n+2)/[\sqrt{2n+3}\ln\varphi]$:

$$\frac{2D_0^2}{n(n-1)\varepsilon_0(1 + \varphi_{\text{var}})(1 + C\varepsilon_0)} = e$$

This is a quadratic equation in ε_0 :

ε_0 from K20

$$C(1 + \varphi_{\text{var}})e\varepsilon_0^2 + (1 + \varphi_{\text{var}})e\varepsilon_0 - \frac{2D_0^2}{n(n-1)} = 0$$

Positive root at $n = 8$:

$$\varepsilon_0 = 0.007550$$

vs $\varepsilon_0^{\text{EIC}} = 0.007550$ (gap 0.005%).

K20 derives $\varepsilon_0 = |I_{\text{eff}} - \ln 2|$ from $n = 8$, Euler's e , and φ alone.

2.2 Physical interpretation

K20 says: the KL-action per wave cycle (K^*T_{avg}) divided by the wave amplitude (F) equals e . The system self-tunes so that the noise floor ε_0 is precisely the value at which this ratio is Euler's constant.

Equivalently: e is the normalisation constant of the CRF wave process. The Born-rule dynamics at $n = 8$ has Euler's number as its characteristic amplitude ratio.

3 K25: Ratio of K20 to K21

K25 — algebraic corollary

$$\frac{K^* \varphi_{\text{var}}}{F \sigma_{\Psi}} = \frac{e}{\sqrt{\varphi}}$$

Numerically: $0.11750 \times 0.4844 / (1.5378 \times 0.01732) = 2.1369$ vs $e / \sqrt{\varphi} = 2.1370$. Gap: 0.004%.

Proof: Divide K20 ($K^* T_{\text{avg}} / F = e$) by K21 ($T_{\text{avg}} \sigma_{\Psi} / \varphi_{\text{var}} = \sqrt{\varphi}$):

$$\frac{K^* T_{\text{avg}} / F}{T_{\text{avg}} \sigma_{\Psi} / \varphi_{\text{var}}} = \frac{K^* \varphi_{\text{var}}}{F \sigma_{\Psi}} = \frac{e}{\sqrt{\varphi}} \quad \square$$

Exact from K20 and K21 by division.

4 The Two-Layer Pinning Structure

Layer 1 (K21, ε_0 -free):

$$f(n) = \frac{4n^3(1 - e^{-1/n})(n+2)}{(n-1)\sqrt{2n+3}(n^2-2)} = \sqrt{\varphi} \quad \Rightarrow \quad n = 8$$

The golden ratio $\sqrt{\varphi}$ selects $n = 8$ uniquely from the $\text{SO}(3)$ mode structure.

Layer 2 (K20, $n = 8$ given):

$$C(1 + \varphi_{\text{var}})e \varepsilon_0^2 + (1 + \varphi_{\text{var}})e \varepsilon_0 - \frac{2D_0^2}{n(n-1)} = 0 \quad \Rightarrow \quad \varepsilon_0 = 0.007550$$

Euler's e selects ε_0 uniquely given $n = 8$.

Together: $(\varphi, e) \rightarrow (n, \varepsilon_0) \rightarrow \text{everything}$.

The two CRF primitives are determined by the two most fundamental mathematical constants: the golden ratio and Euler's number.

5 Complete New Status

ID	Identity	Gap	Status	Note
K20	$K^*T_{\text{avg}}/F = e$	0.005%	Hypothesis	Derives ε_0
K21	$T_{\text{avg}}\sigma_{\Psi}/\varphi_{\text{var}} = \sqrt{\varphi}$	0.004%	Derived	ε_0 -free; pins $n = 8$
K22	$H_u\beta/H_b = e$	0.041%	Hypothesis	Derives H_u
K23	$H_a\beta/\varphi = \sqrt{6}$	0.044%	Hypothesis	$\sqrt{6}$ -web
K24	$n\psi_{\text{grad}}/K^* = \sqrt{6}$	0.046%	Hypothesis	$\sqrt{6}$ -web
K25	$K^*\varphi_{\text{var}}/(F\sigma_{\Psi}) = e/\sqrt{\varphi}$	0.004%	Derived	$K20 \div K21$
	K21 ε_0 -cancellation	exact	Proved	New structural
	K20 derives $\varepsilon_0 = 0.007550$	0.005%	Hypothesis	Quadratic in ε_0
	$(\varphi, e) \rightarrow (n, \varepsilon_0)$	0.005%	Structural	Two-layer pinning

6 Open Questions

K21: Why is $f(8) = \sqrt{\varphi}$ exactly?

$f(n) = \sqrt{\varphi}$ has its unique real solution at $n = 8.00045$ rather than exactly $n = 8$. The 0.004% gap is unexplained. Is there an algebraic identity showing $f(8) = \sqrt{\varphi}$ exactly? This would require:

$$\frac{4 \times 8^3 \times (1 - e^{-1/8}) \times 10}{7 \times \sqrt{19} \times 62} = \sqrt{\varphi}$$

equivalently $5120(1 - e^{-1/8}) = 7\sqrt{19}\varphi$. Numerically LHS = $5120 \times 0.11750 = 601.60$, RHS = $7\sqrt{30.71} = 38.79...$ checking: $5120 \times (1 - e^{-1/8})/(7\sqrt{19}) = 5120 \times 0.11750/30.59 = 19.65$; $\sqrt{\varphi} = 1.272$. The equation is more subtle; the formal proof remains open.

K20: Why Euler's e ?

The Born-resampling dynamics at $n = 8$ tunes itself so that $K^*T_{\text{avg}}/F = e$. Why e specifically? The physical argument (“the amplitude equals the natural growth rate per cycle”) is suggestive but not yet derived from the δ -chain. Formal proof would show that the KL first-passage process at $n = 8$ has e as its characteristic amplitude-to-action ratio.

*K21 carries no ε_0 . It is geometry asking:
 “What n gives $\sqrt{\varphi}$?”
 The answer is $n = 8$.*

K20 then asks: “What ε_0 gives e ?”

The answer is 0.007550.

Two questions. Two constants. Two primitives.

The rest of CRF follows.